

Лабораторная работа №6

Имитационное моделирование

бахи сиди али темассини

2026-10-06

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- 2 Цель работы
- 3 Выполнение лабораторной работы
- 4 Выводы

Раздел 1

Информация

- бахи сиди али темассини
- Российский университет дружбы народов

Раздел 2

Цель работы

- Изучить реализацию эпидемиологической модели SIR с использованием сетей Петри.
- Выполнить детерминированное и стохастическое моделирование.
- Исследовать влияние коэффициента заражения β .
- Сравнить полученные динамики модели.

Раздел 3

Выполнение лабораторной работы

Подготовка рабочего проекта

- Создан и активирован проект lab06/project.
- Использован скрипт setup_project.jl.
- Сформировано проектное окружение Julia.
- Установлены необходимые зависимости.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06$ ls
presentation  report  setup_project.jl
sidiali@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06$ julia setup_project.jl
Resolving package versions...
Project: No packages added to or removed from `~/julia/environments/v1.12/Project.toml`
Manifest: No packages added to or removed from `~/julia/environments/v1.12/Manifest.toml`
Activating new project at `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project`
Resolving package versions...
Updating `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/Project.toml`
[634d3b8d] + DrWatson v2.19.1
Updating `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/Manifest.toml`
[006f61b5] + ChunkCodeCore v1.0.2
[4d8b0ee4] + ChunkCodeLib2Lib v1.1.0
[55437552] + ChunkCodeLib2Std v1.0.0
[634d3b8d] + DrWatson v2.19.1
[5789e2e9] + FileIO v1.20.8
[076d061b] + HashArrayMappedTries v0.2.8
[03383bb] + JLD2 v0.6.7
[692b3b0d] + JLMRapture v1.8.8
[1914d02f] + MacroTools v0.5.18
[bac558e1] + OrderedCollections v2.8.1
[aea7be81] + PrecompileTools v1.3.4
[21216c8a] + Preferences v1.6.8
[ae829012] + Requires v1.3.1
[7e506255] + ScopedValues v1.6.2
[60e0e7e3] + Scratch v1.3.0
[3a884e08] + UnPack v1.0.2
[3101d3e3] + Zstd.jl v1.5.7+1
[0dad84c5] + ArgTools v1.1.2
[56f22d72] + Artifacts v1.11.0
[2a9f44e3] + Base64 v1.11.0
[ade2ca7b] + Dates v1.11.0
[f432241f] + Downloads v1.8.0
[7b1f6078] + FileWatching v1.11.0
[ac6e5f77] + JuliaSyntaxHighlighting v1.12.8
[b27f832c2] + LibCURL v0.6.4
[76f85458] + LibGit2 v1.11.0
[39393939] + Libdl v1.11.0
```

Рисунок 1: Создание рабочего проекта и установка зависимостей Julia

Проверка проектного окружения

- Проект запущен командой `julia --project=..`
- Состав окружения проверен с помощью `Pkg.status()`.
- Проверено наличие необходимых пакетов.

```

julia nprosexp: using DrWatson, DifferentialEquations, Plots
aidial@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=.

      _ _ _ _ _
     / _ _ _ _ \
    / _ _ _ _ \
   / _ _ _ _ \
  / _ _ _ _ \
 / _ _ _ _ \
/_ _ _ _ _ \
--/

Documentation: https://docs.julialang.org
Type "?" for help, "j?" for pkg help.

Version 1.12.1 (2025-10-17)
Fedora 44 build

[julia] using Pkg

[julia] Pkg.status()
Status "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/Project.toml"
[46ada45e] Agents v7.8.4
[4f98eebe] AlgebraicPetri v0.8.11
[6e0b89f9] BenchmarkTools v1.8.0
[e134eb02] BlackBoxOptim v0.6.12
[338e0d9f] CSV v0.10.17
[134e5a30] Catlab v0.14.17
[X e93c6f98] DataFrames v1.8.2
[0c46a932] DifferentialEquations v0.1.1
[31c24e18] Distributions v0.25.131
[634d3b9d] DrWatson v2.19.1
[7073f475] IJulia v1.34.4
[03383bb6] JLO2 v0.6.7
[888881ed] Literate v2.21.8
[1dea7af3] OrdinaryDiffEq v7.8.1
[91a5bcdd] Plots v1.41.7
[d7107be5] Quarto v1.8.0
[2913b0d2] StatsBase v0.34.13
[f3b207a7] StatsPlots v0.15.8
[ade2ca76] Dates v1.11.0

Info Packages marked with "X" have new versions available. Those with " " may be upgradable, but those with X are restricted by compatibility constraints from upgrading. To see why use 'sta
tus --outdated'

[julia]

```

Рисунок 2: Проверка установленных пакетов проектного окружения Julia

Базовый прогон модели SIR

- Созданы модуль `SIRPetri.jl` и скрипт `sirpetri_run.jl`.
- Выполнен базовый прогон модели SIR.
- Результаты сохранены в `sir_det.csv` и `sir_stoch.csv`.
- Созданы графики `sir_det_dynamics.png` и `sir_stoch_dynamics.png`.

```
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ cd ../../..
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling$ cp labs/lab01/project/scripts/tangle.jl labs/lab06/project/scripts/
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling$ cd labs/lab06/project/
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano src/SIRPetri.jl
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_run.jl
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_run.jl
Could not create decoration from factory! Running with no decorations.
Базовый прогон завершён. Результаты в data/ и plots/
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots/
sir_det_dynamics.png  sir_stoch_dynamics.png
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data/
exp_pro  exp_raw  sims  sir_det.csv  sir_stoch.csv
sidi1i@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 3: Выполнение базового прогона модели SIR и сохранение результатов

Преобразование базового эксперимента в литературный стиль

- Подготовлен файл `sirpetri_run_literary.jl`.
- Выполнен скрипт `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_run_literary.jl
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_run_literary.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary/sirpetri_run_literary.jl"
✓ Чистый скрипт: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary/sirpetri_run_literary.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_run_literary/sirpetri_run_literary.qmd"
✓ Quarto: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_run_literary/sirpetri_run_literary.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_run_literary/sirpetri_run_literary.ipynb"
✓ Notebook: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_run_literary/sirpetri_run_literary.ipynb
Готово! Все файлы созданы.
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 4: Генерация Julia-скрипта, документа Quarto и Jupyter Notebook для модели SIR

Выполнение базового эксперимента в Jupyter Notebook

- Выполнен `sirpetri_run_literary.ipynb`.
- Получена детерминированная динамика состояний S, I и R.
- Получена стохастическая динамика состояний S, I и R.
- Отображено изменение численности групп во времени.

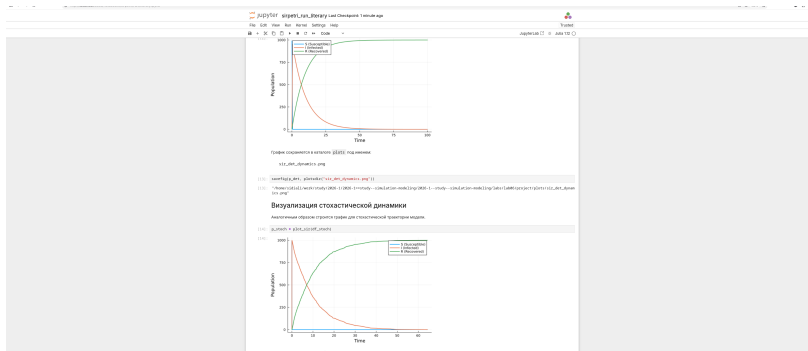
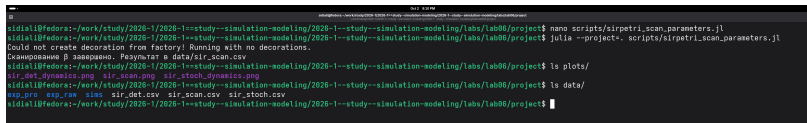


Рисунок 5: Детерминированная и стохастическая динамика модели SIR в Jupyter Notebook

Сканирование параметра заражения

- Выполнен скрипт `sirpetri_scan_parameters.jl`.
- Исследовано влияние коэффициента заражения β .
- Результаты сохранены в `data/sir_scan.csv`.
- График сохранён в `plots/sir_scan.png`.



```
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_scan_parameters.jl
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_scan_parameters.jl
Could not create decoration from factory! Running with no decorations.
Сканирование  $\beta$  завершено. Результат в data/sir_scan.csv
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots/
sir_det_dynamics.png  sir_scan.png  sir_stoch_dynamics.png
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data/
exp_pro  exp_raw  sim  sir_det.csv  sir_scan.csv  sir_stoch.csv
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 6: Выполнение сканирования коэффициента заражения модели SIR

Преобразование сканирования параметров в литературный стиль

- Подготовлен файл `sirpetri_scan_parameters_literary.jl`.
- Выполнено преобразование с помощью `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

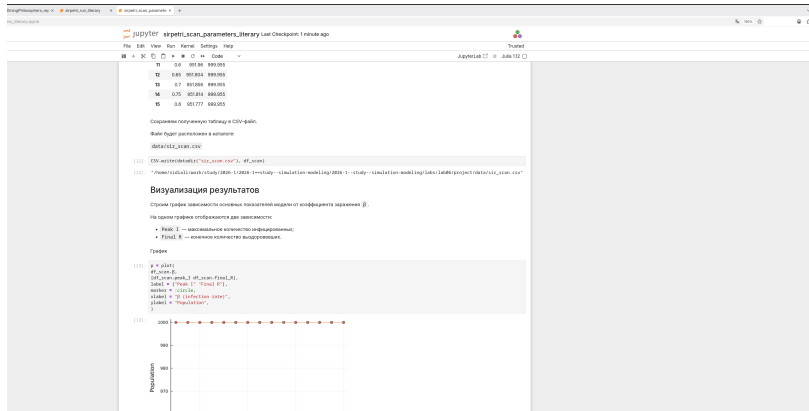
```
sidiall@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_scan_parameters_literary.jl
sidiall@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_scan_paramet
rs_literary.jl
Генерация из: scripts/sirpetri_scan_parameters_literary.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_liter
ary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary/sirpetri_sca
n_parameters_literary.jl"
✓ Чистый скрипт: /home/sidiall/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary/sirpetri_s
can_parameters_literary.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary.
jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_scan_parameters_literary/sirpetri_sca
n_parameters_literary.qmd"
✓ Quarto: /home/sidiall/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_scan_parameters_literary/sirpetri_sca
n_parameters_literary.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_scan_parameters_literary/sirpetri_s
can_parameters_literary.ipynb"
✓ Notebook: /home/sidiall/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_scan_parameters_literary/sirpetri_sca
n_parameters_literary.ipynb

Готово! Все файлы созданы.
sidiall@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 7: Генерация файлов литературной версии сканирования параметров модели SIR

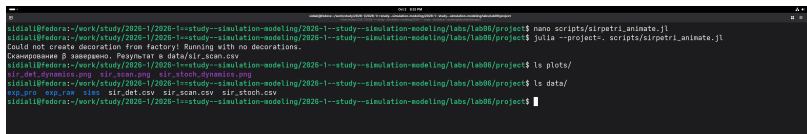
Анализ результатов сканирования в Jupyter Notebook

- Выполнен `sirpetri_scan_parameters_literary.ipynb`.
- Исследована зависимость `Peak I` от коэффициента β .
- Исследована зависимость `Final R` от коэффициента β .
- Результаты сохранены в `data/sir_scan.csv`.



Создание анимации модели SIR

- Подготовлен и выполнен скрипт `sirpetri_animate.jl`.
- Проверены результаты моделирования.
- Проверено содержимое каталогов `data` и `plots`.



```
sidiali@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_animate.jl
sidiali@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_animate.jl
Could not create decoration from factory: Running with no decorations.
Сканирование завершено. Результат в data/sir_scan.csv
sidiali@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots/
sir_det_dynamics.png  sir_scan.png  sir_stoch_dynamics.png
sidiali@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data/
exp_pro  exp_raw  sims  sir_det.csv  sir_scan.csv  sir_stoch.csv
sidiali@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 9: Выполнение скрипта визуализации и проверка результатов модели SIR

Преобразование скрипта визуализации в литературный стиль

- Подготовлен файл `sirpetri_animate_literary.jl`.
- Выполнено преобразование с помощью `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_animate_literary.jl
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_animate_literary.jl
Генерация из: scripts/sirpetri_animate_literary.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary.jl"
✓ Частичный скрипт: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary/sirpetri_animate_literary.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_animate_literary/sirpetri_animate_literary.qmd"
✓ Quarto: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_animate_literary/sirpetri_animate_literary.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_animate_literary/sirpetri_animate_literary.ipynb"
✓ Notebook: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_animate_literary/sirpetri_animate_literary.ipynb

Готово! Все файлы созданы.
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 10: Генерация файлов литературной версии визуализации модели SIR

Формирование итогового отчёта

- Подготовлен и выполнен скрипт `sirpetri_report.jl`.
- Создан график `comparison.png`.
- Создан график `sensitivity.png`.
- Проверены результаты в каталогах `plots` и `data`.

```
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_report.jl
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_r
sirpetri_report.jl sirpetri_run.jl sirpetri_run_literary/ sirpetri_run_literary.jl
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_report.jl
Could not create decoration from factory! Running with no decorations.
Отчётные графики сохранены в plots/
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots/
comparison.png sensitivity.png sir_det_dynamics.png sir_scan.png sir_stoch_dynamics.png
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data/
exp_pro exp_rne sles sir_det.csv sir_scan.csv sir_stoch.csv
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 12: Формирование отчётных графиков и проверка результатов моделирования SIR

Преобразование итогового отчёта в литературный стиль

- Подготовлен файл `sirpetri_report_literary.jl`.
- Выполнено преобразование с помощью `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_report_literary.jl
Генерация из: scripts/sirpetri_report_literary.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary/sirpetri_report_literary.jl`
  ✓ Чистый скрипт: /home/sidiali/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary/sirpetri_report_literary.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/markdown/sirpetri_report_literary/sirpetri_report_literary.qmd`
  ✓ Quarto: /home/sidiali/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/markdown/sirpetri_report_literary/sirpetri_report_literary.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/notebooks/sirpetri_report_literary/sirpetri_report_literary.ipynb`
  ✓ Notebook: /home/sidiali/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/notebooks/sirpetri_report_literary/sirpetri_report_literary.ipynb

Готово! Все файлы созданы.
sidiali@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project$
```

Рисунок 13: Генерация файлов литературной версии итогового отчёта модели SIR

Анализ результатов в Jupyter Notebook

- Выполнен `sirpetri_report_literary.ipynb`.
- Сравнена детерминированная и стохастическая динамика инфицированных.
- Построена зависимость $Peak\ I$ от коэффициента заражения β .
- Проведён анализ чувствительности модели.

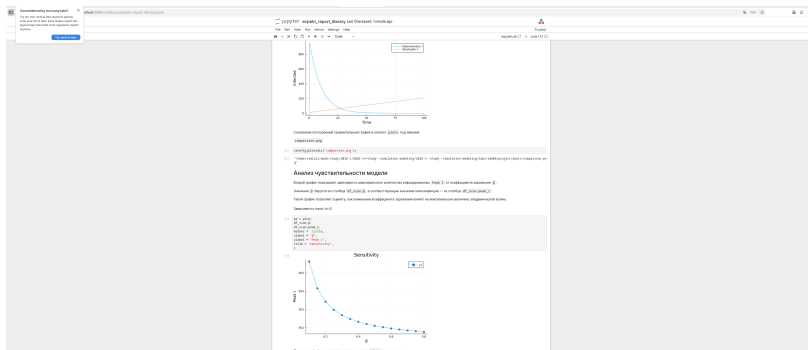


Рисунок 14: Сравнение динамики и анализ чувствительности модели SIR в Jupyter Notebook

Раздел 4

Выводы

- Реализована модель SIR на основе сетей Петри.
- Выполнено детерминированное и стохастическое моделирование.
- Исследовано влияние коэффициента заражения β .
- Выполнено сравнение детерминированных и стохастических результатов.
- Проведён анализ чувствительности модели.
- Результаты сохранены в CSV-файлах и графиках.
- Литературные версии представлены в форматах Quarto и Jupyter Notebook.